FINAL PROJECT REPORT

Predicting USL Championship Match Outcomes Using a Neural Network

Home Win, Draw, or Home Loss from Recent Team Form

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Python + Keras multiclass classification project

Project Snapshot

This report evaluates whether a small neural network can predict the next USL Championship match outcome using only each team's previous five matches. The completed experiment uses the 2024 regular season, avoids future information, and tests the network on the final weeks of the schedule.

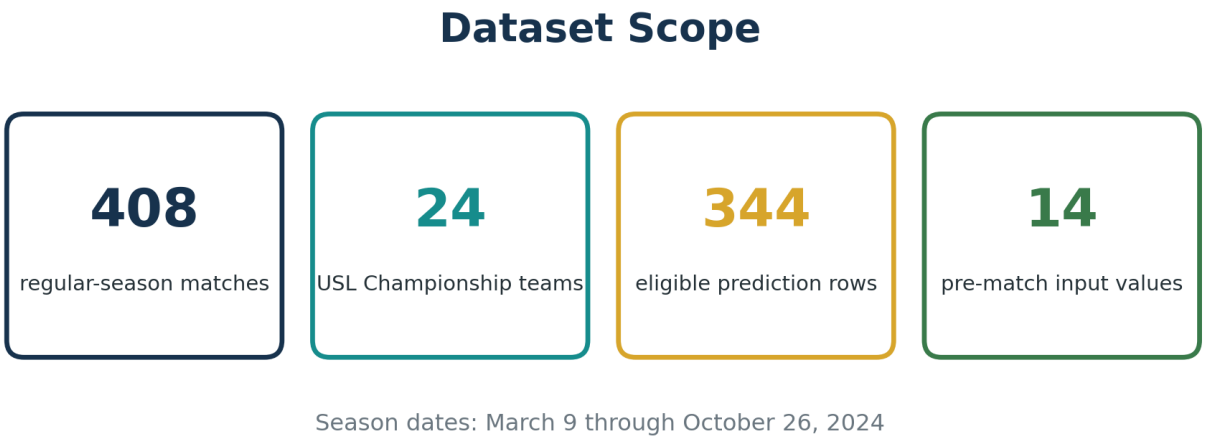


Figure 1. Scope of the raw and prepared match data.

Research question	Can recent five-match form predict a home win, draw, or home loss?
Source	Open USL Championship match-history repository derived from FBref match reports
Prepared rows	344 matches after both teams had at least five earlier matches
Network	14 inputs, Dense 32, Dropout 20%, Dense 16, Softmax 3
Final test	52 matches from October 5 through October 26, 2024
Main result	48.1% accuracy versus 34.6% for always choosing a home win

Honest interpretation

The project produced a modest improvement over a fixed home-win guess, but it did not make USL matches reliably predictable. Home wins were recognized far more often than draws or home losses.

1. Introduction NON-TECHNICAL

Soccer matches are simple to describe but difficult to predict. A team can win, lose, or finish level with its opponent. The final result can turn on one mistake, one strong save, or one unexpected goal. That uncertainty is a major reason people enjoy following a league through an entire season. Supporters often compare recent performances before deciding which club appears stronger. They also think about whether a team is playing at home and whether its attack or defense has been reliable. Even with those clues, confident guesses are often wrong. This project explores how much useful guidance can be found in the patterns teams create over time.

The USL Championship provides a strong setting for this kind of question. Clubs represent communities across the country and meet opponents with different styles and strengths. A team may look excellent one week and struggle against a different matchup the next. Long travel, changing lineups, weather, and the energy of a home crowd can all influence what happens. Supporters still search for patterns because the season gives every club a continuing story. A winning run creates confidence, while a difficult run can make the next match feel uncertain. Looking closely at those patterns makes watching the league more engaging. It also creates a practical way to explore how past events may help people think about the future.

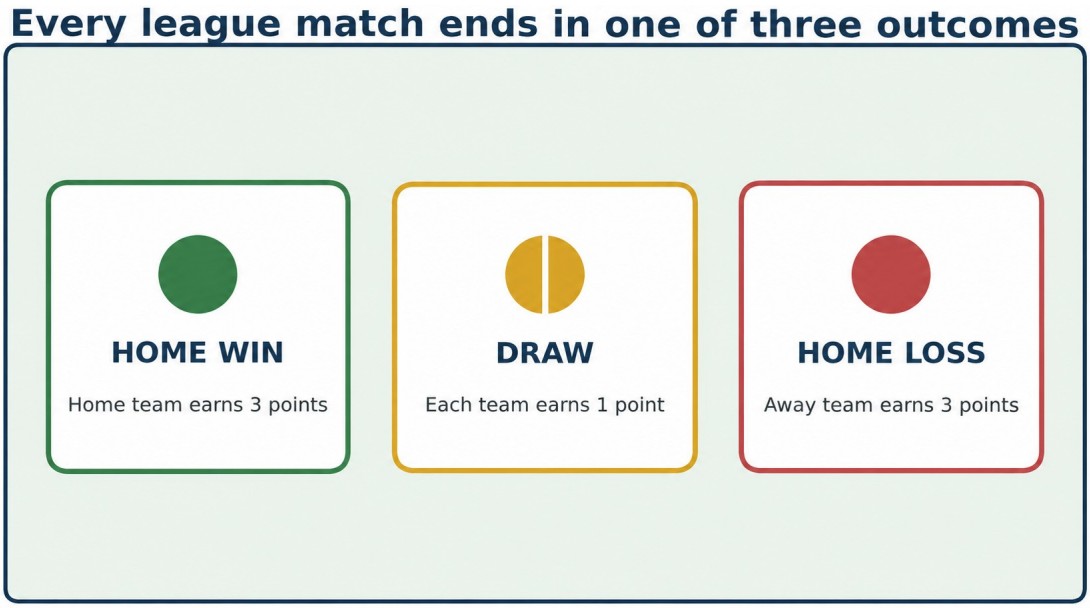


Figure 2. The three possible results from the home team's point of view.

The central question of this project is whether recent team performance can provide a useful estimate of the next result. The goal is not to claim that any match can be known in advance. Instead, the goal is to make a reasonable choice among a home win, a draw, and a home loss. A useful prediction should recognize when the home side appears stronger, when the visitor may have an advantage, and when the matchup looks balanced. It should also admit uncertainty rather than pretending that one answer is guaranteed. A close match should produce less confidence than a matchup with a clear difference in recent form. This makes the project similar to the way an informed supporter might think, but it applies the same rules every time. The final result can then be compared with the estimate to see where the reasoning worked and where it failed.

Match predictions can be useful even when they are far from perfect. A club could use them as one small part of preparing for an opponent. A broadcaster could use them to frame a discussion before kickoff. A supporter could use them to compare an impression with a more consistent estimate. The value comes from organizing information, not from replacing coaches or guaranteeing bets. Soccer includes human decisions and sudden moments that cannot be captured fully before the game begins. Any responsible prediction must therefore be treated as a guide rather than a promise. Understanding those limits is just as important as finding the situations in which the prediction succeeds.

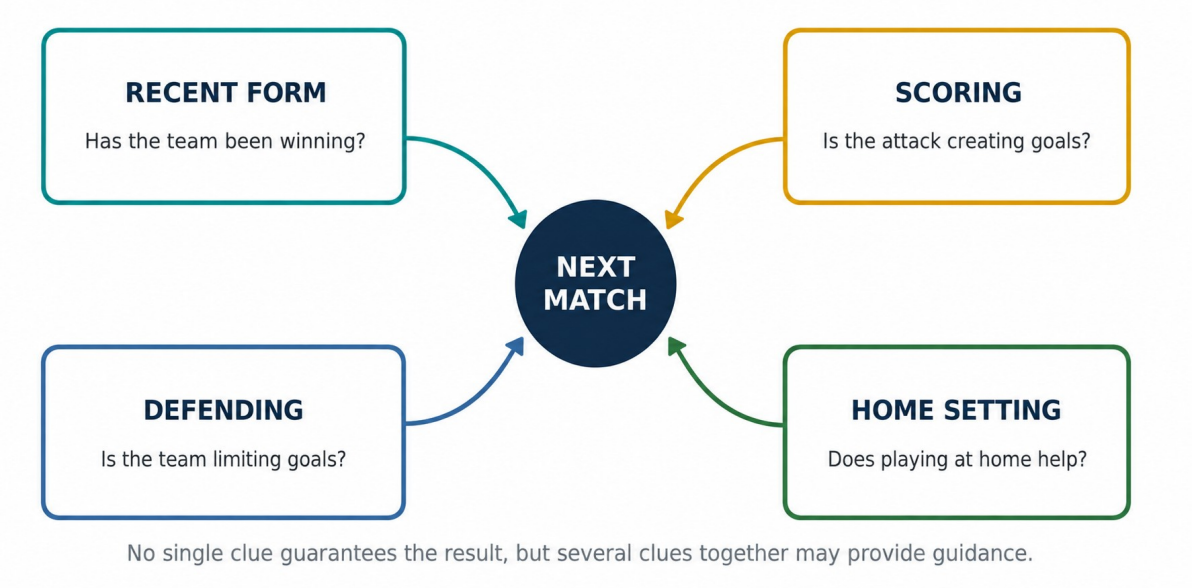


Figure 3. Common nontechnical ideas supporters consider before a match.

This project follows the full story from past matches to future estimates and then to an honest review of the results. It first explains why match outcomes are an interesting problem for ordinary readers. It then shows how recent team patterns can be organized in a fair way without looking ahead. The project asks whether those patterns are strong enough to provide better guidance than a very simple guess. Success is measured not only by the total number of correct choices but also by how well each type of result is recognized. Special attention is given to draws because they often sit between two more decisive outcomes. The findings show both the promise and the difficulty of predicting a sport with so many moving parts. Most importantly, the project demonstrates how a question inspired by following USL soccer can become a complete and testable analysis.

2. Model, Method, and Data Discussion TECHNICAL

2.1 Research Question and Prediction Target

The technical objective was to create a three-class classifier that estimates the outcome of a USL Championship match from the home team's perspective. Class 0 represented a home loss, class 1 represented a draw, and class 2 represented a home win. The output layer returned three probabilities that summed to one, and the class with the largest probability became the predicted outcome.

> **Prediction target**
>
> For each historical match, predict the final home-team outcome using information that existed before kickoff.

2.2 Data Source and Scope

The project was initially planned around the American Soccer Analysis public API. For the completed experiment, a fixed CSV snapshot was used from the open parnold22/usl_championship_data GitHub repository because it provided a reproducible season file that would not change between runs. The repository's match-level records were derived from FBref and included links back to individual match reports. The exact source file was match_level_data_73_2024.csv in the repository's scraped match-statistics directory.

The source covered 408 regular-season matches played by 24 USL Championship teams from March 9 through October 26, 2024. Playoff matches were excluded because extra time and penalty shootouts introduce a different definition of the final result. The project used only the match date, home team, away team, home score, and away score. Attendance, venue, referee, and match-report links were retained in the source but were not used as predictive inputs.

Raw Match Records: Example Rows

Date	Home team	Away team	Home score	Away score	Result
2024-03-09	New Mexico Utd	Pittsburgh	1	0	Home win
2024-03-09	Memphis 901	LV Lights FC	2	1	Home win
2024-03-09	Roots	Indy Eleven	2	1	Home win
2024-03-09	Miami FC	CS Switchbacks	2	0	Home win
2024-03-09	North Carolina	Charleston	0	0	Draw
2024-03-09	Sac Republic	Orange County	2	2	Draw
2024-03-09	Phx Rising	B'ham Legion	0	1	Home loss
2024-03-09	San Antonio FC	Loudoun United	2	2	Draw

Only the date, teams, and final scores were required to construct the prediction dataset.

Figure 4. Example rows from the fixed 2024 match-history file.

Source repository: github.com/parnold22/usl_championship_data
Specific season file: match_level_data_73_2024.csv

2.3 Data Cleaning and Validation

The raw CSV was loaded with pandas so quoted fields such as attendance values with embedded commas stayed in the correct columns. Dates were converted to a chronological date type, the score columns were converted to numbers, rows missing required dates, teams, or scores were removed, and duplicate date-home-away combinations were dropped. The remaining matches were sorted from earliest to latest before any form values were calculated.

A result label was then created from the final score. A home score below the away score became class 0, equal scores became class 1, and a larger home score became class 2. The raw final score was used only to create the label and to update each team's history after the prediction row had been formed. This sequencing was essential because including the current score in its own input would make the prediction meaningless.

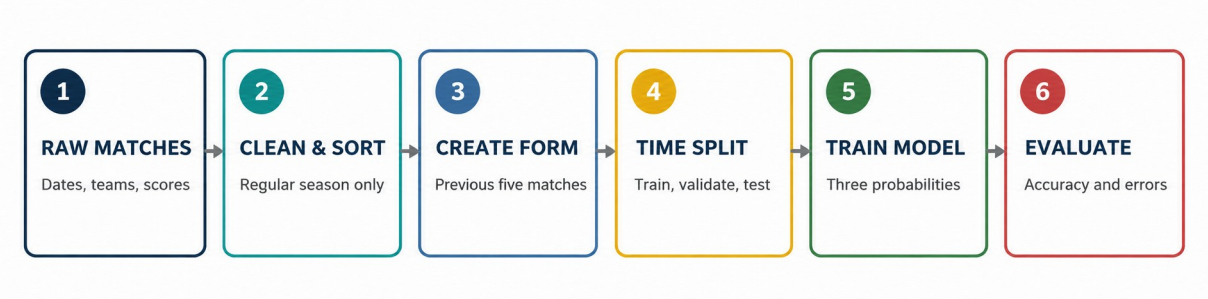


Figure 5. Reproducible path from raw match records to evaluation.

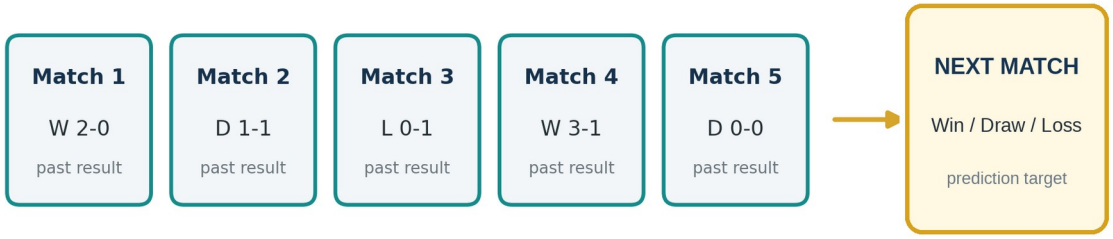
2.4 Feature Engineering and Leakage Prevention

Each prediction row summarized the previous five completed matches for the home team and the away team. Six values were calculated for each club: points per match, goals scored per match, goals allowed per match, win rate, draw rate, and goal difference per match. Two comparison values were added: the difference in points per match and the difference in goal difference. This produced 14 numeric inputs for every eligible match.

Feature group	Number	Examples	Reason for inclusion
Home recent form	6	PPG, GF, GA, win rate, draw rate, goal difference	Describe the home team's last five matches
Away recent form	6	PPG, GF, GA, win rate, draw rate, goal difference	Describe the away team's last five matches
Relative strength	2	PPG difference, goal-difference difference	Compare the teams directly

A match was included only after both teams had five earlier matches. This reduced the 408 raw matches to 344 prepared rows, with the first eligible prediction on April 13. The program calculated the feature row before adding the current match to either team's history. As a result, every input represented information that could have been known before kickoff, which prevented target leakage.

Five-Match History Used Before Each Prediction



The target match never contributes to its own input values. This prevents data leakage.

Figure 6. The five-match rolling window is completed before the target match is added.

Prepared Pre-Match Features: Example Rows

Date	Home	Away	Home PPG	Away PPG	Home GF	Away GF	PPG diff.	Class
2024-04-13	El Paso	Roots	0.20	0.80	0.60	0.60	-0.60	Loss
2024-04-13	Indy Eleven	Charleston	0.80	2.20	1.60	2.00	-1.40	Loss
2024-04-13	Monterey Bay	LV Lights FC	1.60	1.80	1.20	1.40	-0.20	Win
2024-04-13	San Antonio FC	Orange County	1.60	2.20	1.60	1.80	-0.60	Draw
2024-04-20	TB Rowdies	El Paso	1.80	0.20	2.40	1.00	1.60	Draw
2024-04-20	Phx Rising	Pittsburgh	1.40	0.40	1.00	0.20	1.00	Loss
2024-04-20	Orange County	Sac Republic	2.20	2.20	1.40	1.20	0.00	Loss

PPG = points per match; GF = goals scored per match. Every value was calculated from earlier matches only.

Figure 7. Example rows after recent-form features and outcome labels were created.

2.5 Chronological Split and Standardization

The prepared rows were not shuffled. The oldest 70% were used for training, the next 15% were used for validation, and the newest 15% were reserved for the final test. This produced 240 training rows, 52 validation rows, and 52 test rows. The final test covered October 5 through October 26, which simulated asking the trained system to face matches that happened later in time.

A StandardScaler was fitted only on the training inputs. It converted each input to a scale centered near zero with comparable variation, which prevented goals, rates, and point differences from being treated differently simply because they used different numeric ranges. The fitted training scaler was then applied unchanged to the validation and test data. Fitting the scaler on all rows would have allowed information from future matches to influence the training transformation.



Figure 8. Older rows trained the network and the newest rows formed the final test.

2.6 Neural Network Architecture

The classifier was implemented in Python with TensorFlow/Keras through the Keras 3 API. The input layer accepted 14 standardized values. The first dense layer contained 32 units with ReLU activation, followed by a dropout layer that temporarily removed 20% of its outputs during training. A second dense layer contained 16 ReLU units, and the final layer contained three softmax outputs for home loss, draw, and home win.

The network was intentionally small because the prepared dataset contained only 344 examples. A much larger architecture would add parameters without adding evidence and would be more likely to memorize the training rows. Dropout was included as a basic form of regularization. The softmax output was appropriate because each match belonged to exactly one of three mutually exclusive classes.

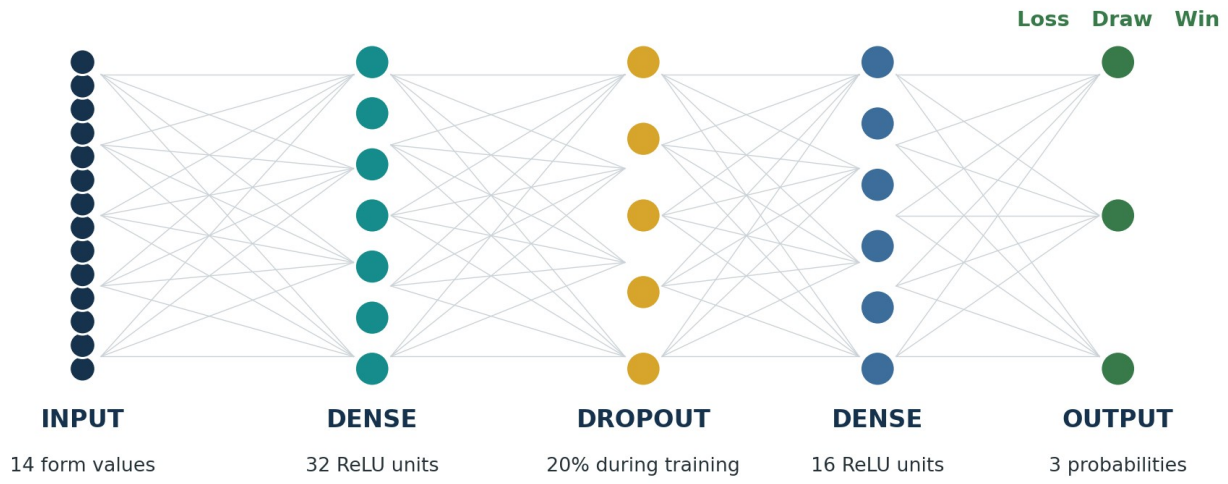


Figure 9. The small Keras network used for three-class prediction.

Layer	Output size	Activation	Purpose
Input	14	None	Receive standardized recent-form features
Dense	32	ReLU	Learn combinations of team-form values
Dropout	32	20% removed in training	Reduce overfitting
Dense	16	ReLU	Create a smaller learned representation
Output	3	Softmax	Return loss, draw, and win probabilities

2.7 Training Procedure

The network used the Adam optimizer with a learning rate of 0.002 and sparse categorical cross-entropy as the loss function. Training used batches of 16 rows and allowed a maximum of 150 epochs. Early stopping monitored validation loss with a patience of 15 epochs and restored the best weights after training. With the fixed random seed of 42, training stopped after 21 epochs.

Accuracy measured the overall portion of correct class selections. Because the three outcomes were not equally common, the evaluation also included precision, recall, and F1 score for each class. A confusion matrix showed exactly which outcomes were being confused. Finally, the network was compared with a fixed baseline that always selected a home win, the most common result in the prepared dataset.

Python, pandas, NumPy, scikit-learn, Keras 3, and Matplotlib were used for data preparation, training, evaluation, and visual reporting.

3. Technical Results and Accuracy Reporting TECHNICAL

3.1 Prepared Data and Outcome Balance

The prepared dataset contained 344 eligible matches. Home wins were the largest class with 156 rows, home losses appeared in 103 rows, and draws appeared in 85 rows. The percentages were 45.3% home wins, 29.9% home losses, and 24.7% draws. This imbalance made accuracy alone incomplete because a system could receive a reasonable-looking score by favoring home wins.

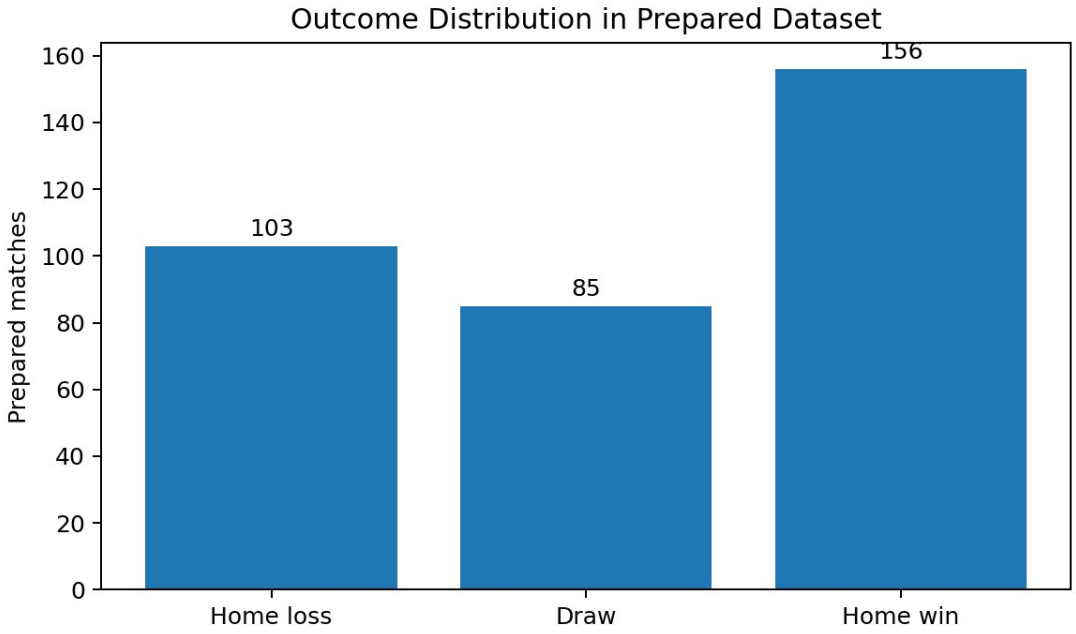


Figure 10. Class distribution in the 344 prepared prediction rows.

3.2 Training and Validation Behavior

Training continued for 21 epochs before early stopping ended the run. Training accuracy generally increased while validation accuracy moved within a narrower range. The best weights were selected according to validation loss rather than the final training epoch. The separation between decreasing training loss and flatter validation loss shows that the network began fitting the training rows more closely than the unseen validation rows.

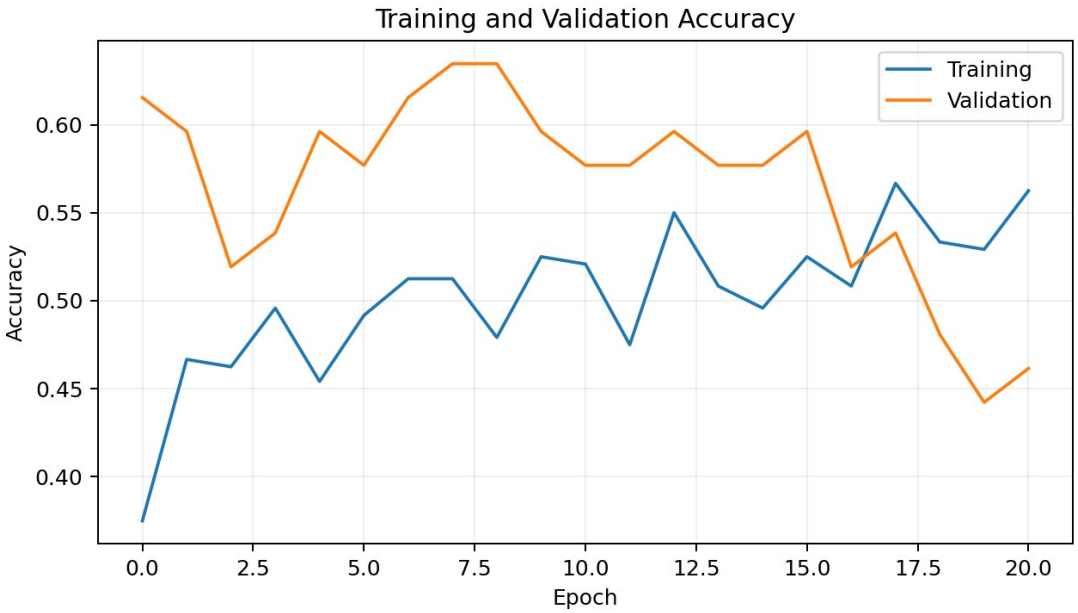


Figure 11. Training and validation accuracy across the completed epochs.

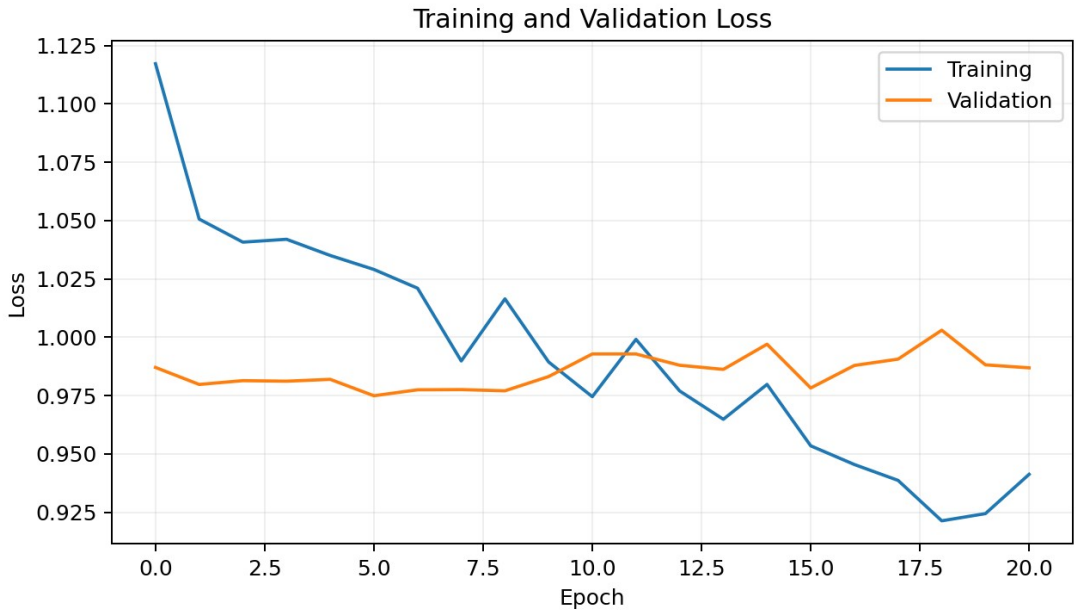


Figure 12. Training and validation loss across the completed epochs.

> **Training interpretation**
>
> The model learned the training rows more easily than the validation rows. Early stopping limited additional overfitting and restored the weights from the best validation-loss point.

3.3 Final Test Accuracy and Baseline

The final test contained 52 matches from October 5 through October 26. The network correctly classified 25 matches, producing 48.1% accuracy. A fixed baseline that selected a home win for all 52 matches would have been correct 18 times, or 34.6%. The network therefore produced seven additional correct outcomes and improved accuracy by 13.5 percentage points over that simple baseline.

The improvement was meaningful for a first experiment, but the absolute accuracy remained limited. A 48.1% result means that 27 of the 52 matches were still classified incorrectly. The model should not be treated as a betting system or a dependable forecast of any individual game. Its value is that it establishes a transparent baseline for testing richer USL information in later work.

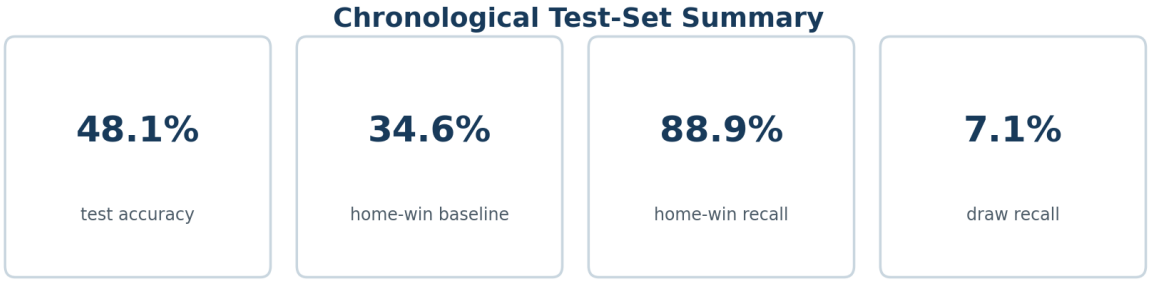


Figure 13. Main measurements from the chronological test set.

Measurement	Value	Interpretation
Test accuracy	48.1%	25 correct predictions out of 52
Always-home-win baseline	34.6%	18 correct predictions out of 52
Accuracy improvement	+13.5 percentage points	Seven additional correct results
Macro F1	40.4%	Average class balance remained modest

3.4 Confusion Matrix

The confusion matrix showed a strong tendency toward predicting home wins. Of the 18 actual home wins, 16 were classified correctly and two were called draws. Of the 20 home losses, eight were classified correctly and 12 were called home wins. Of the 14 actual draws, one was correct, two were called home losses, and 11 were called home wins.

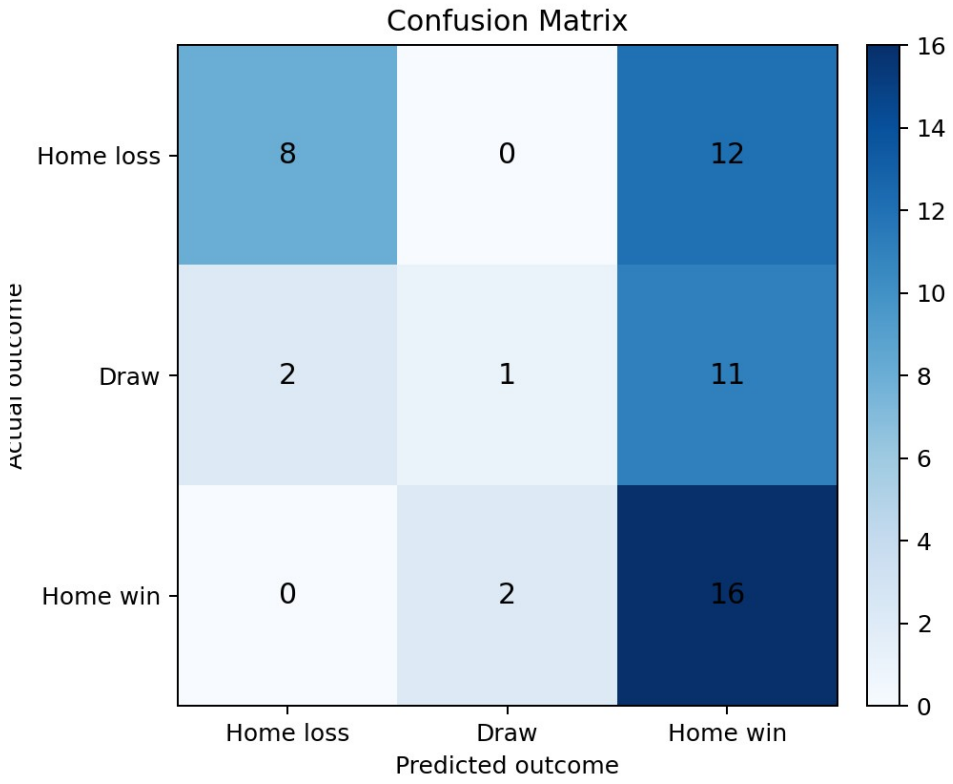


Figure 14. Confusion matrix for the 52 final test matches.

Actual / predicted	Home loss	Draw	Home win
Home loss	8	0	12
Draw	2	1	11
Home win	0	2	16

3.5 Precision, Recall, and F1 by Outcome

Home losses had 80.0% precision, 40.0% recall, and a 53.3% F1 score. Draws had 33.3% precision, 7.1% recall, and an 11.8% F1 score, making them the hardest class by a wide margin. Home wins had 41.0% precision, 88.9% recall, and a 56.1% F1 score, making them the class the network recognized most consistently.

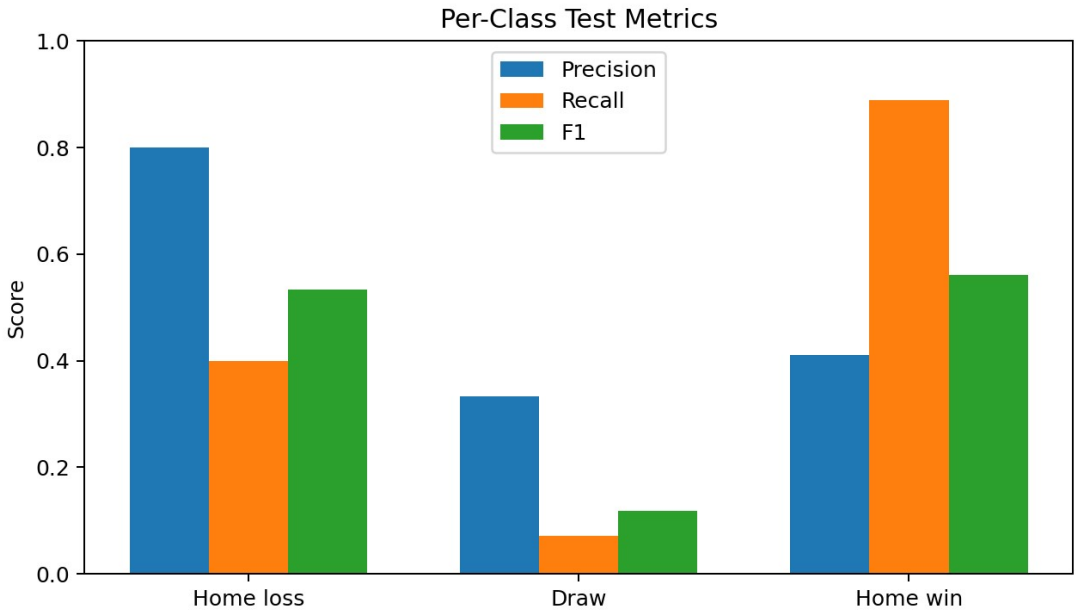


Figure 15. Precision, recall, and F1 score for each test-set outcome.

Outcome	Precision	Recall	F1	Test matches
Home loss	80.0%	40.0%	53.3%	20
Draw	33.3%	7.1%	11.8%	14
Home win	41.0%	88.9%	56.1%	18

3.6 Probability Outputs and Example Predictions

Softmax produced a probability for all three outcomes rather than only a final label. The last six test matches illustrate that the largest probability was not always dominant. The model correctly favored home wins for FC Tulsa, Orange County, and Rhode Island FC on the final date. It also incorrectly favored home wins for Roots, Sacramento Republic, and San Antonio FC when those home teams lost, demonstrating that the model's confidence should be interpreted with caution.

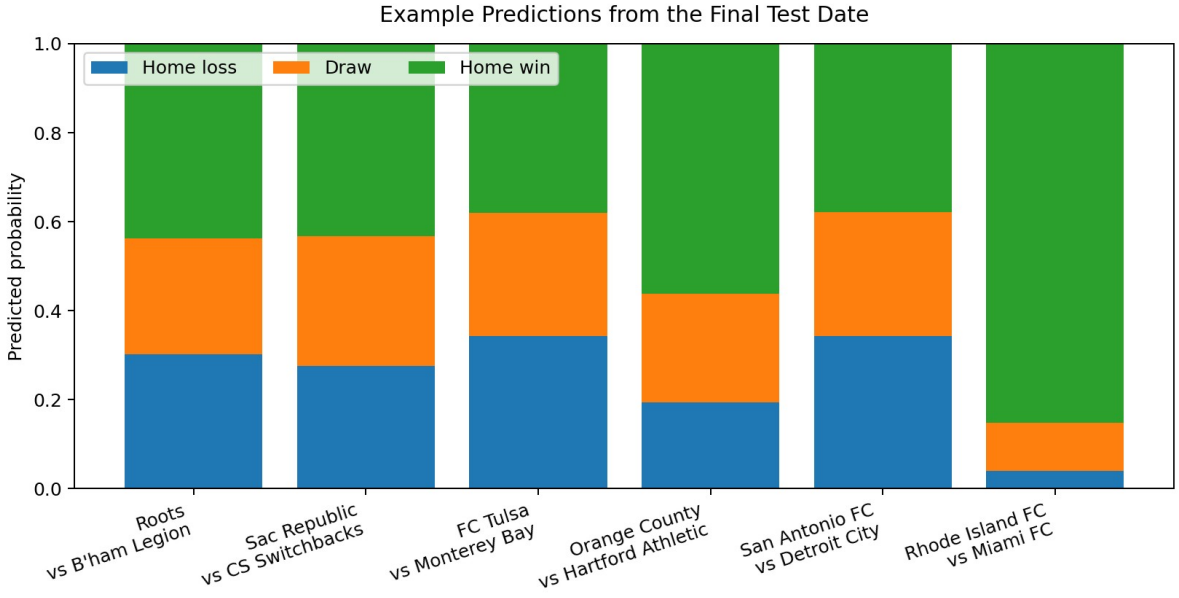


Figure 16. Example loss, draw, and win probabilities from October 26 test matches.

3.7 Limitations and Future Technical Improvements

The largest limitation was the amount and depth of information. The model used only one regular season and only score-based form from the previous five matches. It did not include player availability, starting lineups, travel distance, rest days, coaching changes, home and away splits, expected goals, shot quality, betting markets, or team-strength ratings. These omitted variables can affect a match even when recent final scores look similar.

The chronological test also covered only 52 late-season matches, so a small number of unusual results could move the final accuracy noticeably. Future work should combine several seasons, test multiple lookback lengths, add expected-goal and Goals Added information from American Soccer Analysis, and compare the neural network against logistic regression and tree-based models. Draw prediction should receive special attention through richer chance-quality features or a separate draw-focused decision stage.

> **Most important limitation**
>
> Recent final scores summarize what happened, but they do not explain how it happened. Two teams can have similar five-match records while creating very different chances and facing very different opponents.

4. Non-Technical Conclusions NON-TECHNICAL

This project asked whether recent USL Championship results could help forecast whether the home team would win, draw, or lose. The answer was partly yes, but not strongly enough to make match outcomes feel predictable. The final system chose the correct result in 25 of 52 late-season matches. That was seven more correct choices than simply selecting a home win every time. The improvement shows that recent performance contained some useful information. At the same time, more than half of the matches were still missed. The project therefore produced a useful starting point rather than a dependable forecasting tool. That balance between progress and uncertainty is the main lesson of the work.

The clearest strength was recognizing many of the matches that ended in a home win. It correctly identified 16 of the 18 home wins in the final group of matches. This suggests that strong recent performance and the home setting sometimes create a visible advantage. Home losses were much harder, and eight of twenty were identified correctly. The system often expected the home team to recover even when the visitor ultimately won. That pattern shows how easy it is to give too much weight to the home side. A better version would need a clearer picture of the visiting team's strength. Even so, the home-win results show that the project found at least one meaningful pattern.



Figure 17. The three main take-home messages from the project.

Draws were the most difficult result by a wide margin. Only one of the fourteen draws was called correctly. A draw can happen in a low-scoring defensive match or in a wide-open game where both teams score several times. Those very different paths lead to the same final outcome. Recent wins and goals alone may not describe when two teams are likely to cancel each other out. Information about lineups, playing style, injuries, and expected scoring chances could help in the future. The difficulty with draws also explains why soccer prediction remains challenging even for more advanced systems. In this project, the draw results were the clearest reminder that a simple history cannot tell the whole story.

The project also showed why a fair test matters. Later matches were kept separate from earlier matches so the system could not learn from events that had not happened yet. This made the final test harder, but it made the result more believable. Team quality and form change across a season, especially near the playoff race. A method that looks excellent when old and new matches are mixed together may fail when asked to face the future. The final score should therefore be viewed as an honest estimate of how this simple approach traveled through time. That honesty is more valuable than presenting an inflated number. It gives a clear base that future improvements can be compared against.

Several practical improvements could make a future version stronger. More seasons would provide a much larger collection of examples. Player availability, travel distance, rest time, and home attendance could add context that the recent results did not include. Expected-goal information could describe performance more deeply than the final score alone. Separate measurements of home and away form might also reveal important differences. The current project still achieved its main purpose by turning a soccer question into a complete, testable process. It showed where recent form helps, where it fails, and why uncertainty should remain visible. For a first USL prediction project, that is a useful and honest conclusion.

References

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Appendix A. Feature Dictionary

The following table lists every numeric input sent to the network. All values were calculated from the five completed matches immediately before the target match.

Feature	Definition	Range or unit
home_ppg / away_ppg	Average points earned per match	0 to 3
home_gf / away_gf	Average goals scored	Goals per match
home_ga / away_ga	Average goals allowed	Goals per match
home_win_rate / away_win_rate	Portion of the five matches won	0 to 1
home_draw_rate / away_draw_rate	Portion of the five matches drawn	0 to 1

Feature	Definition	Range or unit
home_goal_diff / away_goal_diff	Average goals scored minus goals allowed	Goals per match
ppg_difference	Home PPG minus away PPG	Points per match
goal_diff_difference	Home goal difference minus away goal difference	Goals per match